



Online

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Regression

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Regression

What is simple linear regression?

- Regression
 - determining the relationship between the value of
 - a **dependent** variableand
 - the values of one or more **independent** variables
- **strength of concrete**
 - as a function of **curing time**
- **working lives** of medical implants
 - as a function of different **surface coatings**
- **transmission times** of data packets
 - as a function of **network load**

What is **simple** linear regression?

- Simple
 - Involves only one **independent** variable

What is simple **linear** regression?

- Linear in terms of *parameters*

The diagram illustrates the components of the simple linear regression equation $Y = \beta_1 X + \epsilon$. The equation is centered, with four labels and arrows pointing to its parts: 'dependent variable' (red) points to Y , 'independent variable' (blue) points to X , 'parameters' (green) points to β_1 , and 'error term' (purple) points to ϵ .

dependent variable

independent variable

$$Y = \beta_1 X + \epsilon$$

parameters

error term

What is simple **linear** regression?

- Linear in terms of *parameters*

The diagram illustrates the components of the simple linear regression equation $Y = \beta_0 + \beta_1 X + \epsilon$. A red arrow points from the label "dependent variable" to Y . A blue arrow points from the label "independent variable" to X . Two green arrows point from the label "parameters" to β_0 and β_1 . A purple arrow points from the label "error term" to ϵ .

dependent variable

independent variable

$$Y = \beta_0 + \beta_1 X + \epsilon$$

parameters

error term

What is simple **linear** regression?

- Linear in terms of *parameters*

The diagram shows the equation $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \epsilon$ with four colored arrows pointing to its parts: a red arrow from 'dependent variable' to Y , a blue arrow from 'independent variable' to X , three green arrows from 'parameters' to β_0 , β_1 , and β_2 , and a purple arrow from 'error term' to ϵ .

dependent variable

independent variable

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \epsilon$$

parameters

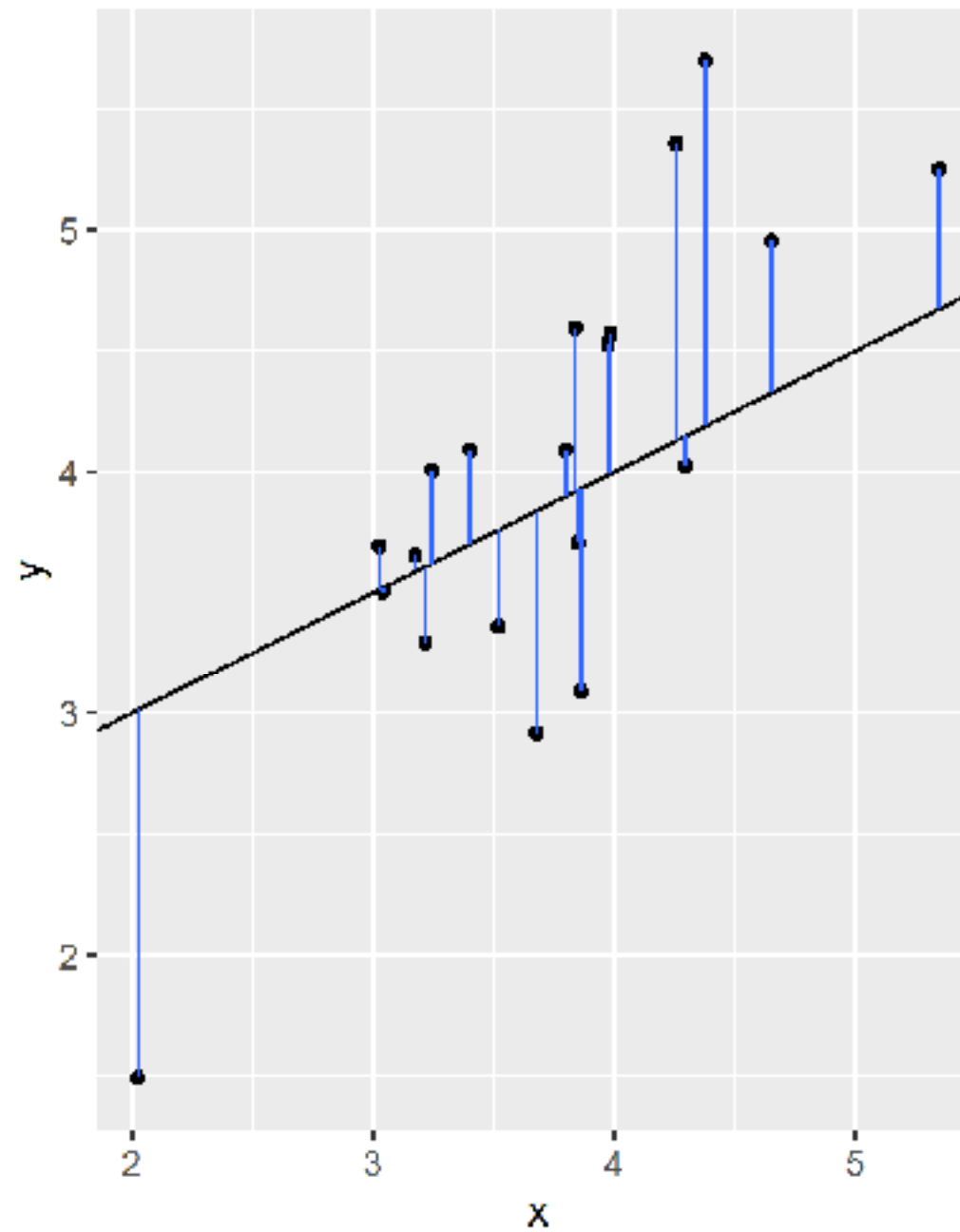
error term

What is simple linear regression?

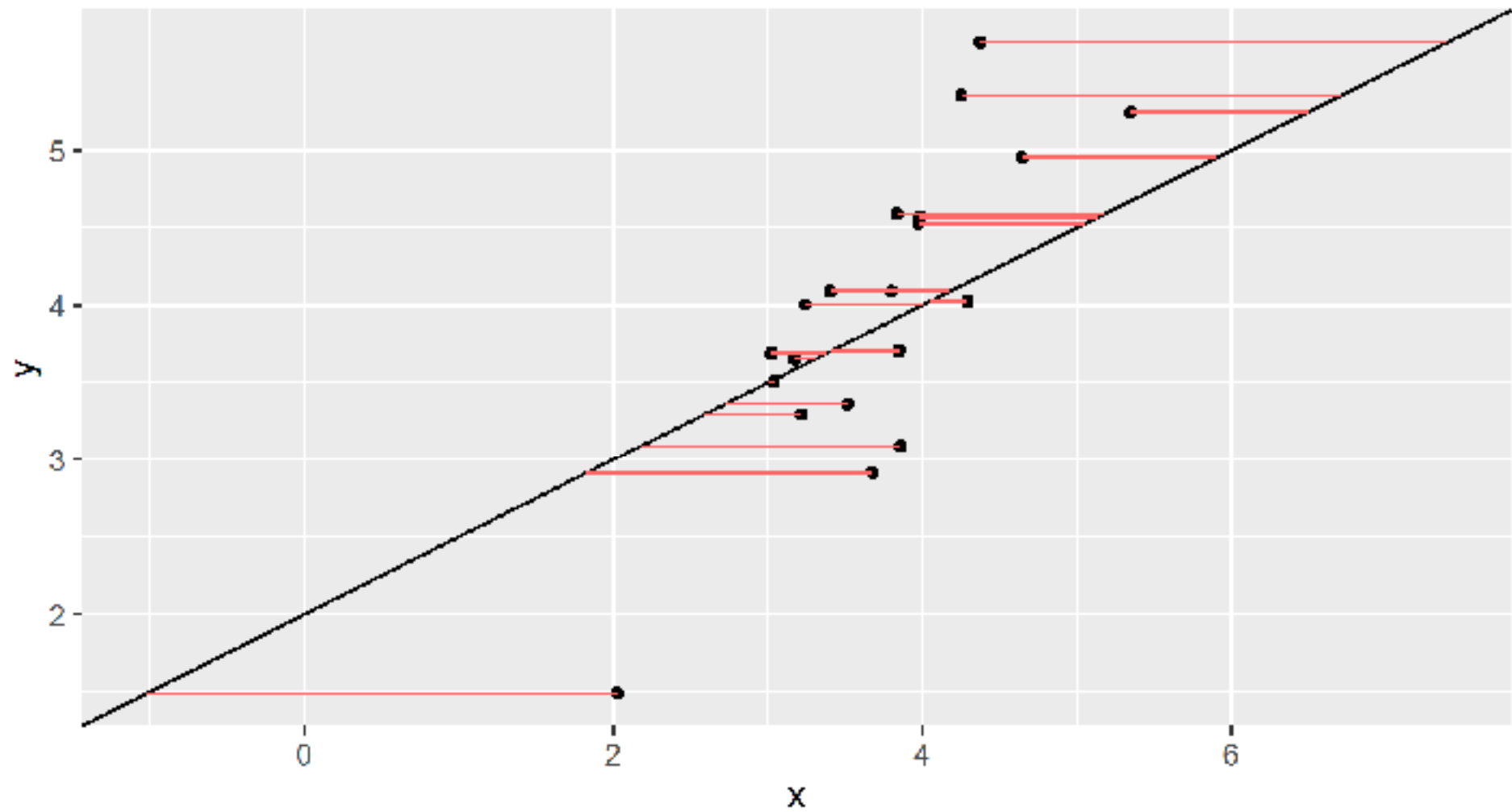
- We will use the simple linear regression model:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

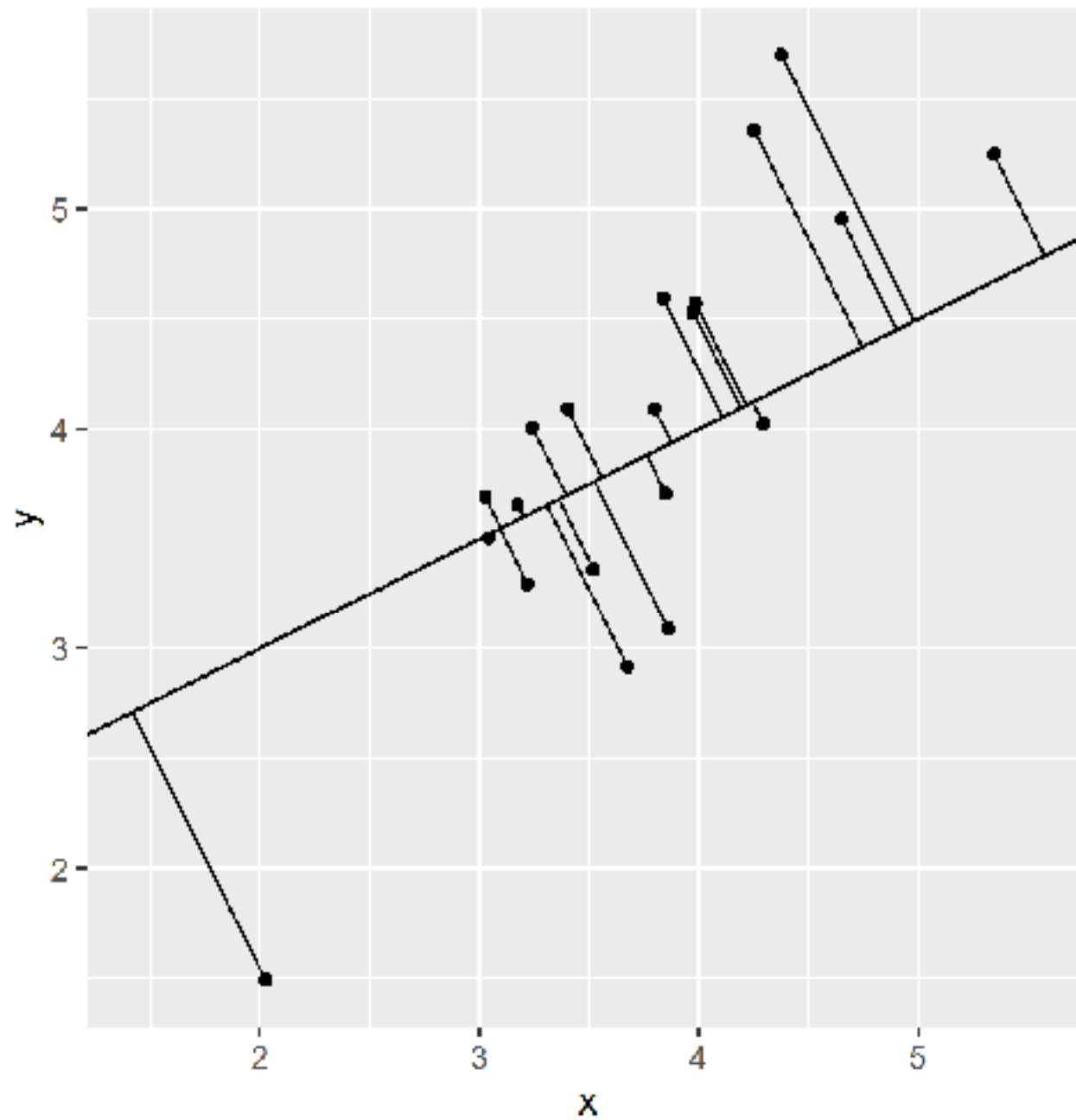
- i.e., fitting a straight line



Ordinary least squares regression: regress y onto x . Used to predict y , given x .
Find the line that minimises the squared y -distances to the data



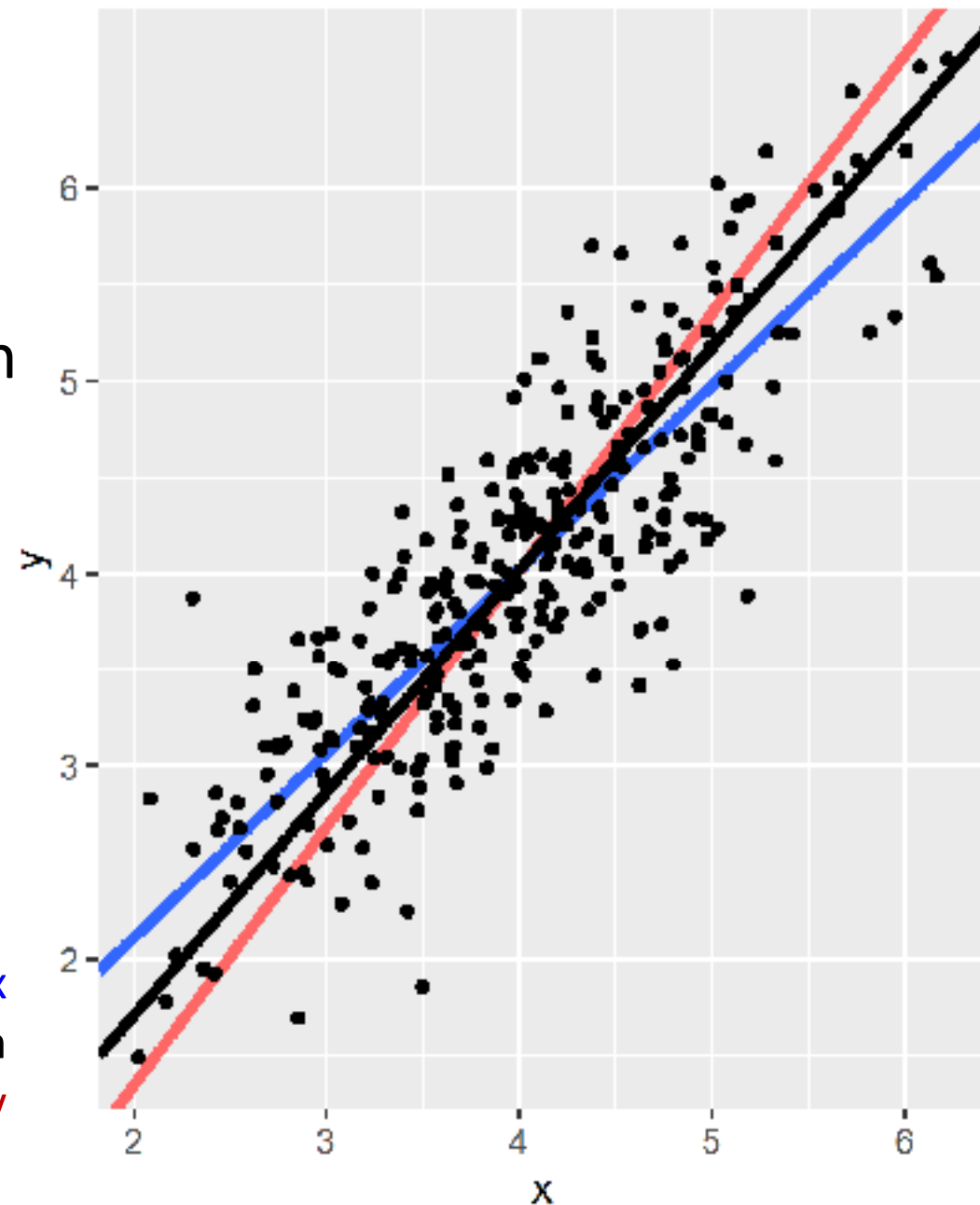
Ordinary least squares regression: regress x onto y . Used to predict x , given y .
Find the line that minimises the squared x -distances to the data



Orthogonal regression. Used when x and y are measured with error. Find the line that gives the best 1-dimensional projection of the data

- Different regression methods yield different lines of “best” fit

y regressed onto x
orthogonal regression
x regressed onto y



Check out [Making sense of principal component analysis, eigenvectors & eigenvalues](#)